ECE637 Project 1: Design and Characterization of a 1-Bit Full Adder

1. Introduction

This project is aiming to design and characterize a 1-bit static CMOS full adder using the TSMC 65-nm technology in Cadence Virtuoso. The objectives are to determine a suitable Pmos width, verify functional correctness, propose transistor sizing to reduce propagation delay, evaluate power-delay performance under process and mismatch variations, and compare pre-layout and post-layout results. Performance is evaluated through transient simulation, process-corner analysis, Monte Carlo simulation, and post-layout simulation.

2. Circuit Design and Transistor Sizing

The full adder accepts three inputs, (A), (B), and (Cin), and produces the sum (S) and carry-out (Cout): Sum=A⊕B⊕Cin and Cout=AB+BC+AC

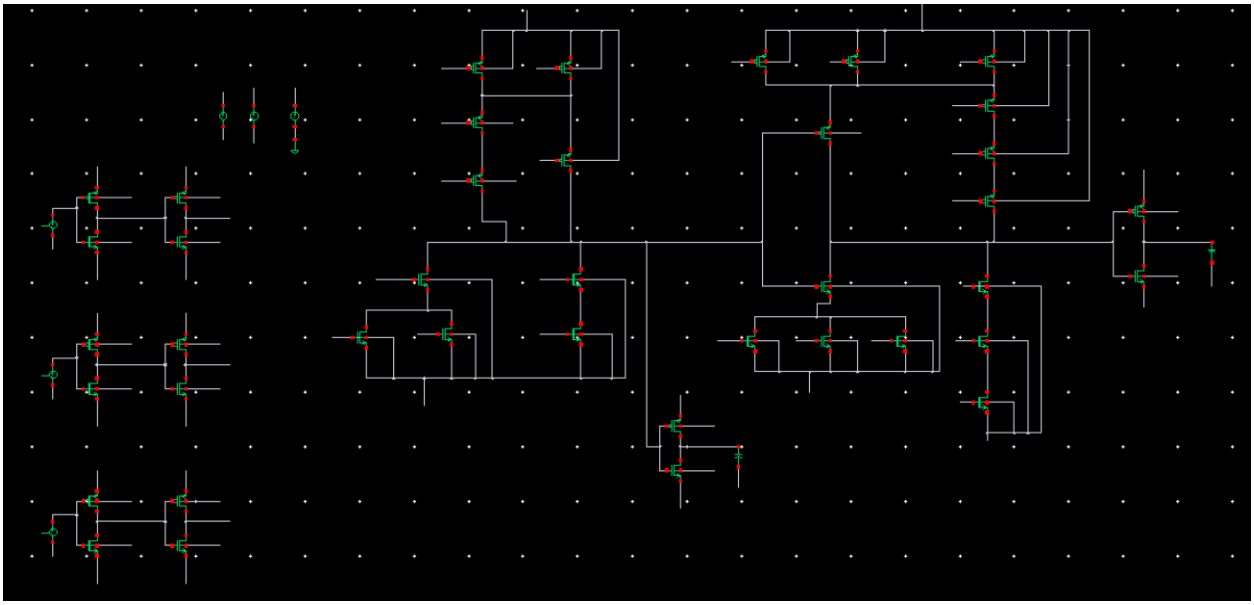


Figure 1. Complete transistor-level schematic of the full adder, including input buffers, power supplies and output loads

A CMOS inverter was first constructed with Wn = 200 nm and L = 60 nm. The PMOS width (Wp) was then swept in 5 nm increments (minimum increase of technology) from 200nm to 600nm. For each width, tpLH and tpHL were measured. The selected width was Wp = 405 nm, as it produced the smallest difference between the two propagation delays. Increasing or decreasing Wp by 5 nm resulted in a larger difference.

Parameters: wp=400n			
3	project1_inv_tb_1	/VIN	
3	project1_inv_tb_1	/VOUT	
3	project1_inv_tb_1	(cross(clip(VT("/...	48.25p
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Parameters: wp=405n			
4	project1_inv_tb_1	/VIN	
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Figure 2. Comparison of tpLH and tpHL showing that Wp = 405 nm yields the smallest difference between the two propagation delays

3. Simulation Results and Analysis

3.1 Functional Waveforms

Functional verification used a Gray-code sequence covering all eight input combinations. The simulation verified the expected truth table.

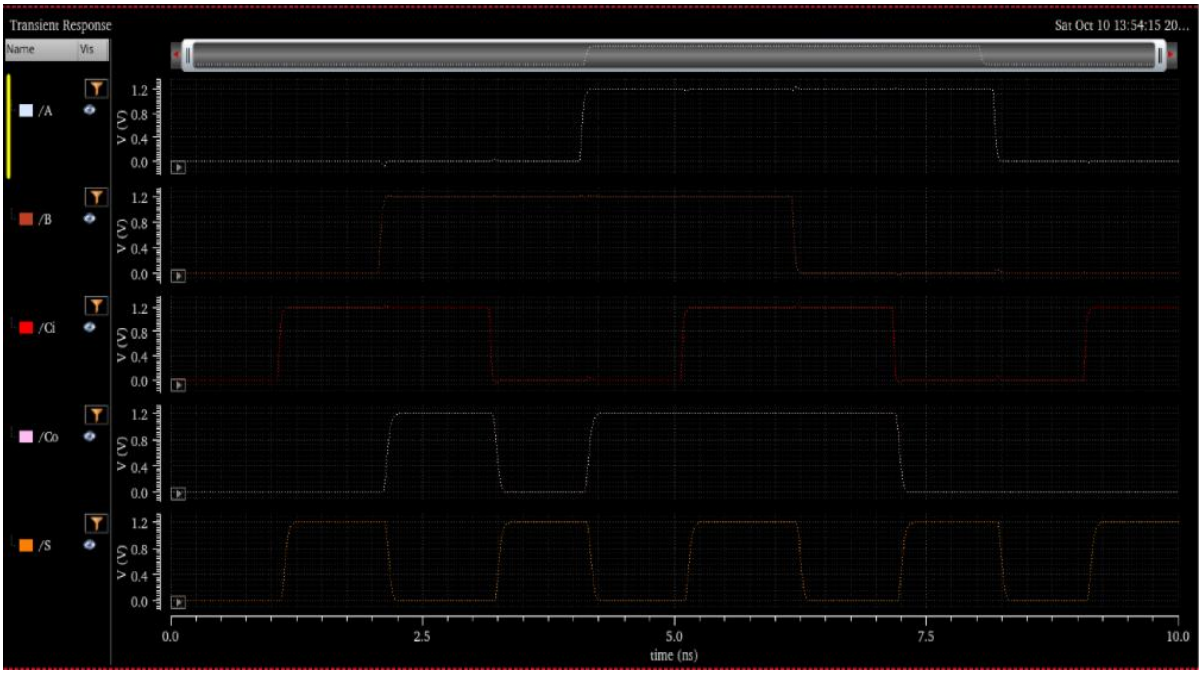


Figure 3. Transient waveforms of A, B, Cin, S, and Cout, showing all eight input combinations

3.2 Worst-Case Input-Transition Combinations and alternative transistor sizing

All 56 directed transitions between distinct input states were tested by setting the initial and final values of A, B, and Cin for each case. Changing inputs used VPULSE sources with a 1.2 V

swing, 100 ps rise/fall times, and a transition time of 5 ns; unchanged inputs were held at DC 0 V or 1.2 V. Delays were measured at the 0.6 V crossing, using the latest-changing input and the later-changing output among those that transitioned using equation td = max(tS, tCout) − max(tA, tB, tCin) (of course td calculation will only include those input/outputs that actually changed during transition) where each value e.g. tS measured individually. Cases with no output transition were marked N/A. The worst case was 110 → 100, with a delay of 0.0998 ns (rounded to 0.1 ns). Here 110 represents VA=VB=1.2V and VCin=0V initially.

Rank	Input transition (A,B,Cin)	Input transition(s)	Output transition(s)	Delay
1	**110 → 100**	B ↓	S ↑, Cout ↓	**0.1 ns**
2	**001 → 110**	A ↑, B ↑, Cin ↓	S ↓, Cout ↑	**0.095 ns**
3	**010 → 110**	A ↑	S ↓, Cout ↑	**0.093 ns**
4	**011 → 100**	A ↑, B ↓, Cin ↓	S ↑, Cout ↓	**0.090 ns**
5	**100 → 110**	B ↑	S ↓, Cout ↑	**0.090 ns**
6	**110 → 010**	A ↓	S ↑, Cout ↓	**0.089 ns**
7	**001 → 011**	B ↑	S ↓, Cout ↑	**0.083 ns**
8	**110 → 001**	A ↓, B ↓, Cin ↑	S ↑, Cout ↓	**0.082 ns**
9	**011 → 010**	Cin ↓	S ↑, Cout ↓	**0.081 ns**
10	**101 → 100**	Cin ↓	S ↑, Cout ↓	**0.081 ns**

Table 1. Top 10 worst-case input transitions

Since tS >= tCout in all cases, the transistor widths in the S logic network were increased by 25% as an alternative sizing strategy. The new widths in the S logic network are Wn' = 250 nm and Wp=506.25nm, rounded to 505nm to match the technology's supported dimensions. The channel length and Cout network remain unchanged. This aims to reduce resistance and improve drive strength, although increased parasitic capacitance may offset the improvement.

The same 56 transitions were simulated again using the new sizing. The new worst-case delay was 0.101 ns, occurring for transition 110 → 100. Compared with the original worst-case delay, the new sizing did not reduce but instead increased the delay.

Rank	Input transition (A,B,Cin)	Input transition(s)	Output transition(s)	Delay
1	**110 → 100**	B ↓	S ↑, Cout ↓	**0.101 ns**
2	**001 → 110**	A ↑, B ↑, Cin ↓	S ↓, Cout ↑	**0.096 ns**
3	**010 → 110**	A ↑	S ↓, Cout ↑	**0.095 ns**
4	**011 → 100**	A ↑, B ↓, Cin ↓	S ↑, Cout ↓	**0.091 ns**
5	**100 → 110**	B ↑	S ↓, Cout ↑	**0.091 ns**
6	**110 → 010**	A ↓	S ↑, Cout ↓	**0.091 ns**
7	**001 → 011**	B ↑	S ↓, Cout ↑	**0.084 ns**
8	**110 → 001**	A ↓, B ↓, Cin ↑	S ↑, Cout ↓	**0.083 ns**
9	**101 → 100**	Cin ↓	S ↑, Cout ↓	**0.083 ns**

10	**011 → 010**	Cin ↓	S ↑, Cout ↓	**0.082 ns**

Table 2. Top 10 worst-case input transitions using the alternative sizing

The alternative sizing slightly worsened the worst-case delay, increasing it by approximately 1.20%. The worst-case transition remained (110 → 100), suggesting that increasing the S-network transistor widths alone was insufficient to improve the critical transition. In fact, the order or the ten cases are also unchanged except for a switch of ordering of top 9 and 10. Although wider transistors can reduce effective drive resistance, they also increase parasitic capacitance, which can offset the expected speed of improvement. The results indicate that the resistance reduction did not outweigh the additional loading effects for this transition. Further optimization should examine the critical path and its individual node delays rather than attempting to increase transistor widths to reduce the overall propagation delay.

3.3 Power, Delay, PDP, and Global/Local Mismatch

Average power was calculated with equation $P_{avg} = (1/T) \int_0^T V_{DD} \cdot I_{DD}(t)\, dt$ where T=10ns. The power-delay product was calculated as PDP = Pavg × td. The power measurement accounted for both the full adder and the second inverter in the two-stage input buffer.

Process Corner	Temperature	Delay (ns)	Average Power (μW)	PDP (fJ)
TT	-25°C	0.094	4.311	0.405
TT	27°C	0.1 (99.8p)	4.032	0.403 (0.402 if used 99.79p)
SS	-25°C	0.1176	3.891	0.458
SS	27°C	0.1246	4.111	0.512
FF	-25°C	0.078	3.699	0.288
FF	27°C	0.0825	4.873	0.402
SF	-25°C	0.092	3.975	0.365

SF	27°C	0.097	4.565	0.443
FS	-25°C	0.098	4.806	0.469
FS	27°C	0.103	3.966	0.409

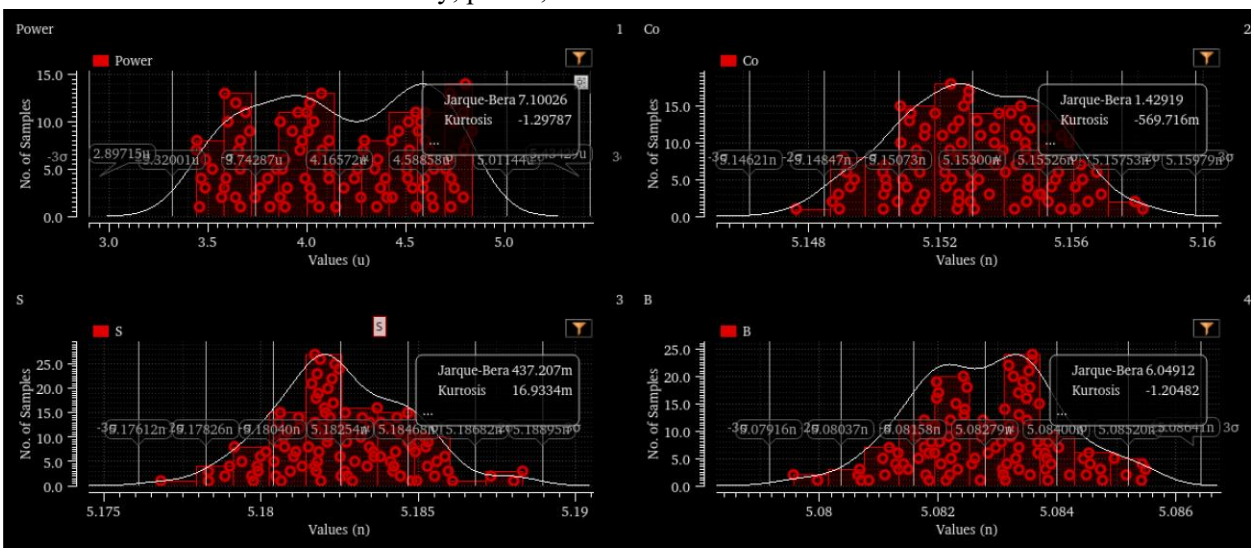
Table 3. Power, delay and PDP for TT, SS, FF, SF, FS at –25 and 27 °C

The SS corner at 27°C represents the worst case for both delay 0.1246ns and PDP 0.512fJ. The worst corner for power is FF at 27°C, with 4.873µW. These results suggest that the SS corner generally produces longer delays due to slower transistor operation, while the FF corner exhibits higher power consumption at 27°C. Increasing the temperature generally increases delay, although its effect on power varies across process corners.

Local mismatch was evaluated using 100 Monte Carlo samples at TT, 27 °C, under the original worst-case input-transition condition.

Test	Name	Yield	Min	Target	Max	Mean	Std Dev	Cpk	Errors
Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter
Yield Estimate: 100 %(100 passed/100 pts)	Confidence Level: <not set>	Filter: <not set>							
project1_tb_adder_mac_1									
	Power	100% (100/100)	3.441u	info	4.837u	4.166u	422.9n		0
	Co	100% (100/100)	5.148n	info	5.158n	5.153n	2.264p		0
	S	100% (100/100)	5.177n	info	5.188n	5.183n	2.139p		0
	B	100% (100/100)	5.08n	info	5.085n	5.083n	1.209p		0
	delay	100% (100/100)	96.04p	info	103.4p	99.75p	1.8p		0

Table 4. Monte Carlo results for delay, power, and PDP



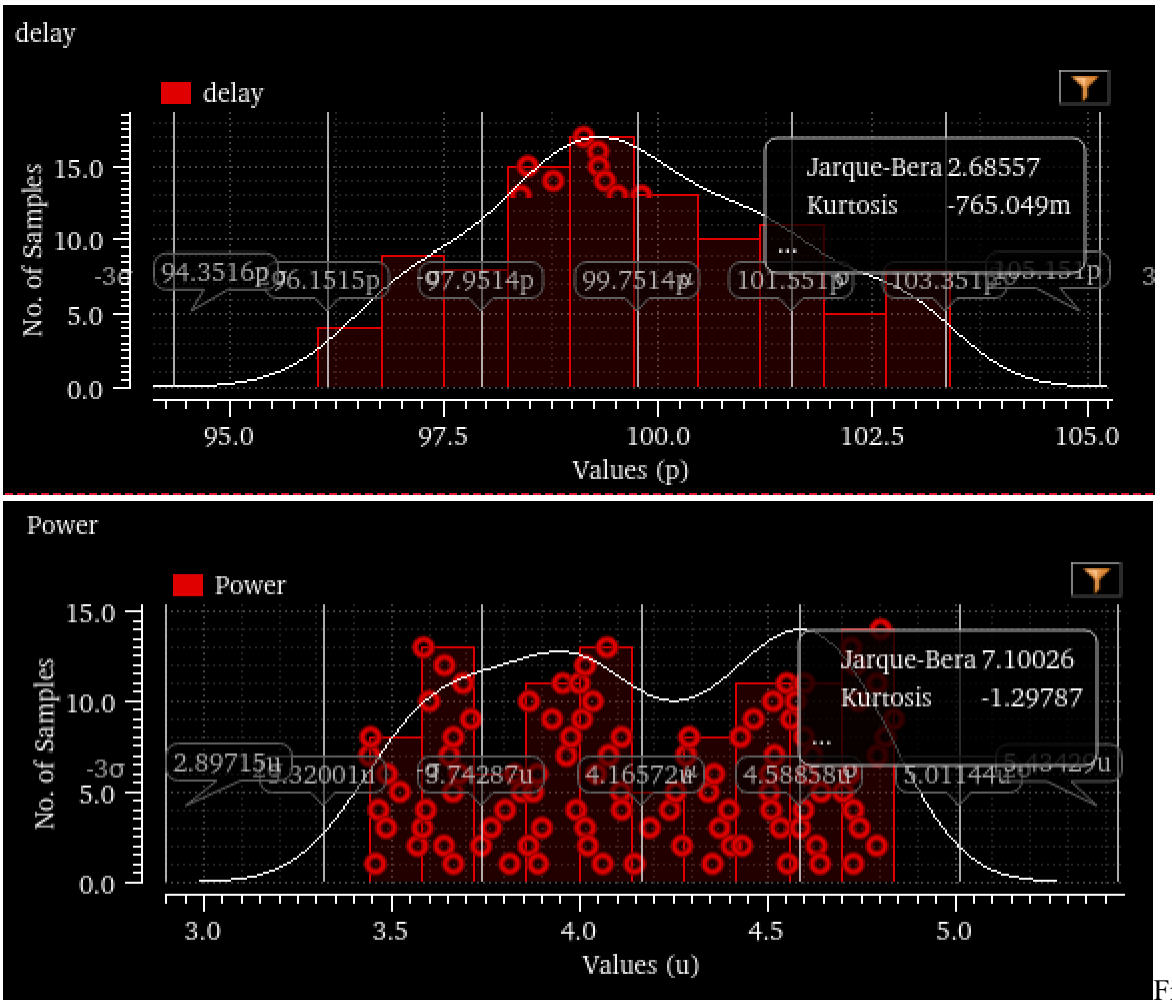


Figure 4. Output graph of the Monte Carlo simulation

Across 100 samples, the measured delay ranged from 96.04 ps to **103.4 ps** (worst-case delay), average power ranged from 3.441 µW to **4.837 µW** (worst-case power). The delay had a mean of 99.75 ps and a standard deviation of 1.8 ps, indicating relatively small sample-to-sample variation. The average power had a mean of 4.166 µW and a standard deviation of 0.4229 µW, showing greater relative variation than delay. The coefficients of variation were approximately 1.80% for delay and 10.15% for power. Hence, the results indicate that the circuit's delay was relatively consistent under local mismatch, whereas its power consumption was more sensitive to sample variation.

4. Layout and Post-Layout Simulation

The 1-bit full-adder layout was completed and verified using Design Rule Check (DRC) and Layout Versus Schematic (LVS). Both verification checks passed without errors, confirming that the layout satisfies the applicable design rules and matches the schematic circuit connectivity.

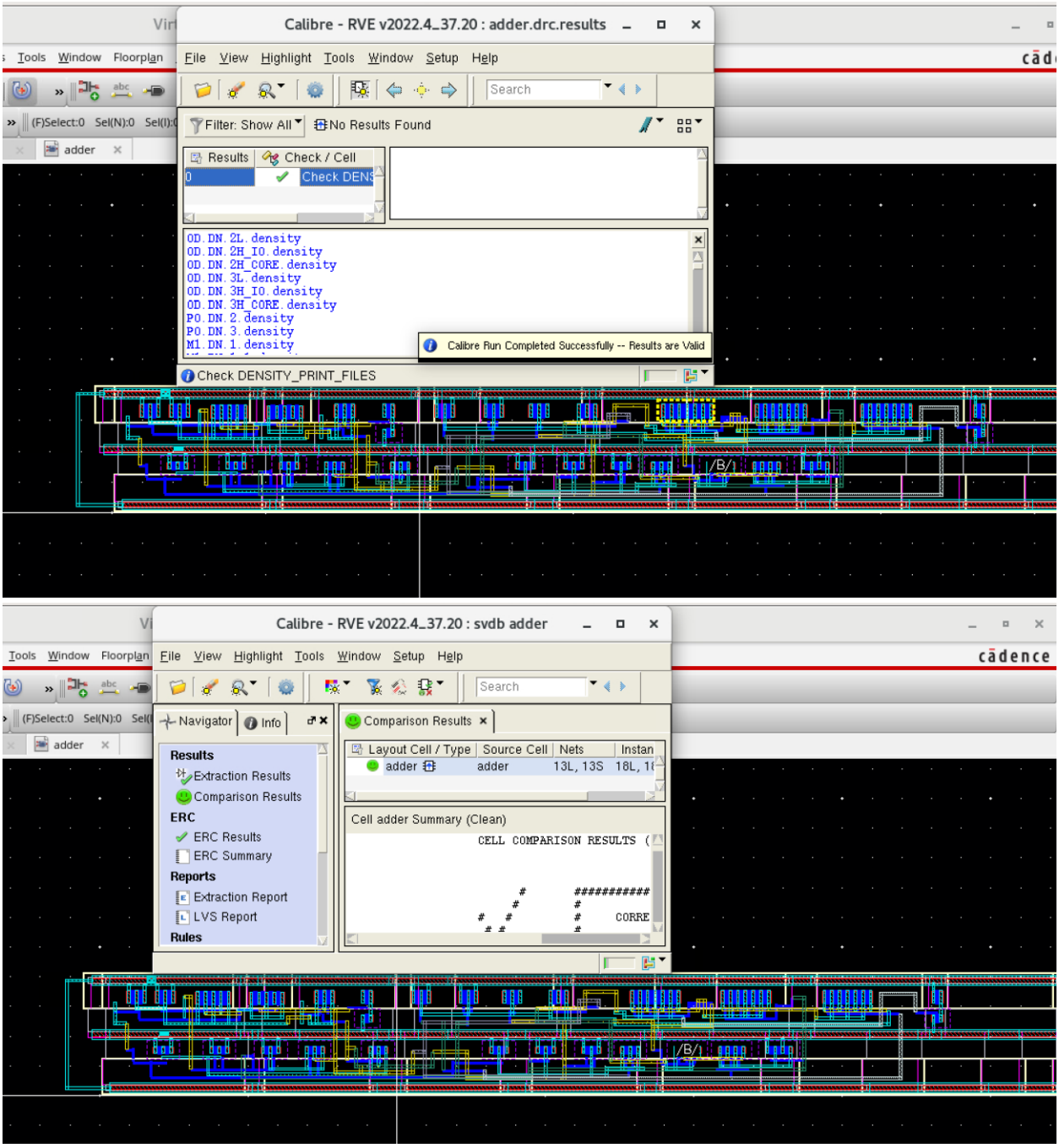


Figure 5. Full-adder layout with error-free DRC and LVS verification results.

Post-layout simulation was performed using Calibre under TT process conditions at 27°C. All 56 cases were simulated again, and the ten worst-case output delays are listed below in table 5.

Rank	Input transition (A,B,Cin)	Input transition(s)	Output transition(s)	Delay
1	**010 → 110**	A ↑	S ↓, Cout ↑	**0.175 ns**
2	**001 → 110**	A ↑, B ↑, Cin ↓	S ↓, Cout ↑	**0.168 ns**

3	**110 → 100**	B ↓	S ↑, Cout ↓	**0.163 ns**
4	**011 → 100**	A ↑, B ↓, Cin ↓	S ↑, Cout ↓	**0.156 ns**
5	**000 → 100**	A ↑	S ↑	**0.152 ns**
6	**110 → 010**	A ↓	S ↑, Cout ↓	**0.151 ns**
7	**100 → 110**	B ↑	S ↓, Cout ↑	**0.150 ns**
8	**111 → 011**	A ↓	S ↓	**0.146 ns**
9	**011 → 111**	A ↑	S ↑	**0.142 ns**
10	**001 → 011**	B ↑	S ↓, Cout ↑	**0.140 ns**

Table 5. Top ten worst-case output delays from post-layout simulation

The worst-case post-layout delay was 0.175 ns, occurring during the input transition from 010 to 110, in which input A rises while B remains high and Cin remains low. Compared with the pre-layout worst-case delay of 0.0998 ns, the post-layout delay increased by 0.0752 ns, corresponding to an increase of approximately 75.35%. This increase is primarily attributed to the parasitic resistance and capacitance introduced by the physical layout, including interconnects, contacts, and device connections. These parasitic effects influence signal propagation and increase propagation delays. Furthermore, the difference in the worst-case input transitions between the pre-layout and post-layout simulations indicates that layout parasitic can affect the relative timing performance of different input combinations, changing which transition produces the maximum delay.

It is also worth noting that the pre-layout and post-layout circuits use different transistor configurations: the pre-layout circuit uses single-finger transistors, whereas the post-layout circuit uses multi-finger transistors. Therefore, the difference in delay rankings may also be partially influenced by finger-related effects. To evaluate their impact, the single-finger and multi-finger implementations were compared using the pre-layout schematic. As an example, case 110 → 100, is shown in Table 6.

Metric	**Single-Finger**	**Multi-Finger**	**Change**
Power	4.032 μW	3.874 μW	−3.92%
Delay	99.79 ps	91.72 ps	−8.09%
PDP	402.4 aJ	355.4 aJ	−11.68%

Table 6. Comparison of single-finger and multi-finger implementations.

The comparison shows that the multi-finger implementation has slightly lower power, delay, and PDP than the single-finger implementation. The multi-finger implementation exhibited an 8.09% reduction in delay, which is substantially smaller than the increase observed after layout. Furthermore, for the same input transition, 110 → 100, the delay increased from 99.8 ps before layout to 163 ps after layout, representing an increase of approximately 63.3%. These results indicate that the extracted layout parasitics are the primary contributor to the delay increase,

while the change in transistor finger configuration has a comparatively smaller effect that can be ignored when comparing the change of delay between pre-layout and post-layout.

5. Conclusion

A 1-bit static CMOS full adder was designed and evaluated in Cadence Virtuoso using TSMC 65-nm technology. Functional simulation verified the expected truth table, and transistor sizing identified a PMOS width of 405 nm that minimized the difference between the inverter's rising and falling propagation delays. Increasing the transistor widths in the sum logic network by 25% slightly worsened the worst-case delay, highlighting the trade-off between reduced transistor resistance and increased capacitive loading. Process-corner and Monte Carlo simulations characterized global and local variations, and post-layout simulation showed the impact of layout parasitic. Overall, the results demonstrate the trade-off between transistor sizing and parasitic effects in optimizing full-adder performance.